05 / REPORT

# The plan, tested.

Read by Gemini / gemini-3.8-flash

## Summary

Before: 100 tests: 1 single weak spot, 3 hidden pairs, 12 already broken by the weak spot, 84 survived.

After: 100 tests: 0 single weak spots, 0 hidden pairs, 0 already broken by a weak spot, 100 survived.

3 fixes accepted: 1 from the AI, 2 yours.

The AI suggested 12 fixes. The checker turned down 10 and couldn't apply 1.

Main human cost before fixes: Routes A, B, C: 30 households not delivered. Nobody can open the loading area

## The plan

AI read 32 items, 1 corrected by you, 0 added by you

### People

Joel is available from 09:00 until 10:30. · From, Until · L3 · Joel can drive. · Driving · L16 · Joel, Maya and Sam hold loading-area keys. · Key · L3 · changed by Fix 2

Ravi is available from 09:00 until 13:00. · Until · L15 · Ravi · mentioned on L8 · Ravi can drive. · Driving · L8 · Ravi does not hold a loading-area key. · Ravi · mentioned on L8

Maya is available from 09:00 until 10:45. · Until · L15 · Maya · mentioned on L8 · Maya can drive. · Driving · L16 · Maya holds a loading-area key. · Maya · mentioned on L8 · changed by Fix 2

Sam is available from 09:00 until 12:00. · Until · L15 · Sam · mentioned on L9 · Sam can drive. · Driving · L9 · Sam holds a loading-area key. · Sam · mentioned on L9 · changed by Fix 2

Anita is available from 09:00 until 12:00. · Until · L15 · Anita · mentioned on L9 · Anita can drive. · Driving · L10 · Anita does not hold a loading-area key. · Anita · mentioned on L9

### Vehicles

Van is available from 09:30. · From · L5

Car is available from 09:00. · From · L5

Loading bay opens at 09:00 and needs a key. · Opens at, Needs a key · L3 · Only one vehicle can load at a time. · One vehicle at a time · L12

### Routes

Route A serves 10 households by car; Ravi drives, with Maya, Joel and Sam as backups. Loaders: Ravi, Joel, Anita and Maya. · Households, Vehicle, Drivers · L8 · Loaders · L13 · changed by Fix 3 · Route A's loading takes 10 minutes. · Loading · L12 · Route A's drive takes 60 minutes. · Driving · L8

Route B serves 12 households by van; Sam drives, with Anita and Ravi as backups. Loaders: Joel, Anita and Maya. · Households, Vehicle, Drivers · L9 · Loaders · L13 · Route B's loading takes 10 minutes. · Loading · L12 · Route B's drive takes 35 minutes. · Driving · L9

Route C serves 8 households by van; Anita drives, with Sam and Ravi as backups. Loaders: Joel, Anita, Maya and Sam. · `Households, Vehicle, Drivers · L10 · Loaders · L13 · changed by Fix 1` · Route C's loading takes 10 minutes. · `Loading · L12` · Route C's drive takes 60 minutes. · `Driving · L10`

### Times

Supply truck drops arrives at 09:00. · `Arrival · L4`

Everything must be delivered by 12:00. · `Finish time · L6`

### Review decisions

- What time is Ravi available from? **Answered: 09:00.**
- What time is Maya available from? **Answered: 09:00.**
- What time is Sam available from? **Answered: 09:00.**
- What time is Anita available from? **Answered: 09:00.**
- Hall has no loading activity or supply arrival listed. Is it separate, or the same place as another listing? **Answered: Hall is the same place as Loading bay. Use Loading bay's reviewed details.**

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## What was tested

`100 tests: every single problem and every pair that can happen. 14 problems, with confirmed sizes.`

| Problem | Size tested |
|---|---|
| Ravi absent | `Absent (no duration)` |
| Joel absent | `Absent (no duration)` |
| Anita absent | `Absent (no duration)` |
| Maya absent | `Absent (no duration)` |
| Sam absent | `Absent (no duration)` |
| Ravi leaves 90 min early | `90 minutes` |
| Joel leaves 90 min early | `90 minutes` |
| Anita leaves 90 min early | `90 minutes` |
| Maya leaves 90 min early | `90 minutes` |
| Sam leaves 90 min early | `90 minutes` |
| Van back 30 min late | `30 minutes` |
| Car back 30 min late | `30 minutes` |
| Supply truck drops 45 min late | `45 minutes` |
| Loading bay blocked 30 min | `30 minutes` |

## What broke before fixes

### ✕ Joel absent

Single weak spot. Routes A, B, C: 30 households not delivered. Nobody can open the loading area

### ✕ Ravi absent + Supply truck drops 45 min late

Hidden pair: each alone survived. Route A: 10 households not delivered

### ✕ Anita absent + Van back 30 min late

Hidden pair: each alone survived. Route C: 8 households not delivered

### ✕ Anita leaves 90 min early + Van back 30 min late

Hidden pair: each alone survived. Route C: 8 households not delivered

12 pairs already broken by Joel absent alone

## Fixes

### Accepted changes

#### ✓ Fix 1 (AI)

Route C: add loader cover: Sam. · `fixes 2 · still broken 2 · new crosses 0`

Suggested by Groq / openai/gpt-oss-120b (fallback: Gemini unavailable)

#### ✓ Fix 2 (yours)

Give Maya a spare loading-area key; existing holders keep theirs. Give Sam a spare loading-area key; existing holders keep theirs. · `fixes 1 · still broken 1 · new crosses 0`

#### ✓ Fix 3 (yours)

Route A: add driver cover: Sam. · `fixes 1 · still broken 0 · new crosses 0`

### Turned-down and couldn't-apply suggestions

Scores below use the plan at each suggestion's point in the history.

Case: Anita leaves 90 min early + Van back 30 min late · Suggested by Groq / openai/gpt-oss-120b (fallback: Gemini unavailable)

#### Extend Anita's availability so she can stay for the loading of Route C.

**Turned down:** This change sets Anita's finish to 11:00, earlier than their planned 12:00, so the case still fails. · `fixes 0 · still broken 4 · new crosses 2`

**Add Ravi as an additional driver for Route C so the route can be driven even if Anita departs early.**

**Turned down:** Ravi is already a driver for Route C. Van back 10:45. Loading team gone. · `fixes 0 · still broken 4 · new crosses 0`

**Assign the Car to Route C (instead of the Van) and give the Van to Route B, avoiding the late-return conflict.**

**Turned down:** Fixes 3 crosses but makes 'Supply truck drops 45 min late' a new single weak spot. · `fixes 3 · still broken 1 · new crosses 1`

Case: Anita absent + Van back 30 min late · Suggested by Groq / openai/gpt-oss-120b (fallback: Gemini unavailable)

**Replace Anita with available drivers for Route C.**

**Turned down:** Creates a new hidden pair: Ravi absent + Sam absent. · `fixes 0 · still broken 4 · new crosses 2`

**Assign the van to Route C earlier and use the car for Route B, preventing the van's late return from blocking Route C.**

**Couldn't apply:** it named the van where a route was needed. · Not scored.

`Unknown target id 'vehicle-van' in fix move_route.`

Case: Joel absent · Suggested by Groq / openai/gpt-oss-120b (fallback: Gemini unavailable)

**Give a spare key to another volunteer so the loading bay can be opened.**

**Turned down:** Creates a new hidden pair: Joel absent + Ravi absent. · `fixes 1 · still broken 1 · new crosses 1`

**Replace Joel as a driver on Route A with another available driver.**

**Turned down:** Creates a new hidden pair: Ravi absent + Maya absent. · `fixes 0 · still broken 2 · new crosses 2`

**Replace Joel as a loader on all routes with another available loader.**

**Turned down:** Creates a new hidden pair: Anita absent + Van back 30 min late. · `fixes 0 · still broken 2 · new crosses 4`

Case: Ravi absent + Supply truck drops 45 min late · Suggested by Groq / openai/gpt-oss-120b (fallback: Gemini unavailable)

**Replace absent driver Ravi with an available driver for Route A.**

**Turned down:** Creates a new hidden pair: Anita absent + Supply truck drops 45 min late. · `fixes 1 · still broken 0 · new crosses 4`

**Give Anita a spare key so she can load the supply truck at the loading bay.**

**Turned down:** Route A is still not delivered. Ravi is absent; the backups leave before the 60-minute drive could finish after the supplies arrive at 9:45. · `fixes 0 · still broken 1 · new crosses 0`

**Add an extra loader (Sam) to Route A to ensure loading can start promptly when the supply truck arrives.**

**Turned down:** Route A is still not delivered. Ravi is absent; the backups leave before the 60-minute drive could finish after the supplies arrive at 9:45. · `fixes 0 · still broken 1 · new crosses 0`

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## After the fixes

✓ `All 100 tested scenarios survive.`

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## What was NOT tested

- Three or more problems at once.
- Problem types outside the list: bad weather, someone getting sick partway through, wrong addresses, running short of food, volunteers arriving late.
- Other sizes of the same problems. Only the sizes listed above were tested.
- Anything the plan did not mention.

**Question-mark results:** none in the original or current run.

Survived means survived the listed problems at the listed sizes. It is not a guarantee.

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This report used a saved AI reading of the example plan. `Fingerprint: 401c1d1d3fc278c5ebdbdf3f5f8e7fe0249aa386ac8a5babffe6803a3dd7e155`